A GUIDED COURSE

Quantum Krylov & Sample-based Diagonalisation

From molecular physics through the qubit Hamiltonian to Ankush’s SKQD

Derive the algorithms / Understand the reported implementation / Design a controlled extension

Manuscript: Ankush Kumar, 8 October 2026 • Expanded companion: 9 October 2026

The learning route



01 First derive the projection. Then identify which basis is actually diagonalised.

02 Finish with replication tests, not just a benchmark table.

How the 13 note chapters become a course

Study phase	Notes chapters	What you will be able to do
Foundations	1–2	Build a molecular target; derive Rayleigh–Ritz.
Coherent quantum Krylov	3–4	Calculate QKS matrices and diagnose stability.
Sample-based method	5–6	Derive S=I; quantify shot-limited discovery.
Ankush’s implementation	7–8	Trace four reductions and interpret hardware claims.
Reproduction + research	9–13	Run labs, compare baselines, defend an extension.

Follow the chapter numbers in the 60-page study notes.

How to read every claim

Source-derived

[M]: uploaded Ankush manuscript, with page or equation.

Reported numbers are preserved, not independently certified.

[P]: pasted explanation; a scaffold, not independent evidence.

Added teaching material

[K], [S], [N], [T]: primary literature.

Derivation / lab: calculations developed here.

Clarification: an implementation input the uploads do not specify.

A molecule becomes an electronic Hamiltonian

$$H_e = -\sum_i \frac{\nabla_i^2}{2} - \sum_{i,A} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + E_{\text{nuc}}$$

01 Water has electron motion, attraction, repulsion and fixed nuclei.

02 Atomic units simplify the constants; the geometry fixes E_{nuc}.

03 This continuous operator is not yet a qubit circuit.

Read each term physically

Electronic contributions

Kinetic: electrons can move.

Electron–nucleus attraction: binds electrons.

Electron–electron repulsion: creates correlation.

Fixed nuclei

Nuclei have a specified geometry.

Their mutual repulsion is E_{nuc}.

Changing the geometry changes the Hamiltonian and energy surface.

A finite basis makes the problem computational



01 An orbital basis transforms integrals into a finite-dimensional model.

02 An active-space choice changes the target approximation; tapering later changes its representation.

Separate chemistry error from algorithm error

Defines the target

Nuclear geometry / clamped nuclei

Finite orbital basis and frozen cores

Active electrons, orbitals, symmetry sector

Solves a given target

Jordan–Wigner: exact encoding when done correctly

Sector tapering: exact within a verified sector

QKS/SKQD: approximate eigenvalue within a trial space

Second quantization uses orbital occupations

$$H = E_{\text{const}} + \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} g_{pqrs} a_p^\dagger a_q^\dagger a_s a_r$$

01 Creation a† adds and annihilation a removes an electron.

02 Fermionic anticommutation enforces the exclusion principle.

03 The integrals h and g are classical numbers from the orbitals.

Jordan–Wigner turns fermions into Pauli operators

$$a_p^\dagger = \left(\prod_{j<p} Z_j\right) \frac{X_p - iY_p}{2}, \qquad n_p = \frac{I - Z_p}{2}$$

01 The Z string enforces the fermionic exchange sign.

02 An empty orbital is |0⟩; an occupied orbital is |1⟩.

03 Collecting mapped terms gives H = Σα cα Pα.

Compute the Jordan–Wigner minus sign

01 **Occupations**

Printed bit order is $b_3 b_2 b_1 b_0$.

$$|0011\rangle : \quad n_0 = n_1 = 1,\ n_2 = n_3 = 0$$

02 **Annihilation**

Remove electron from orbital 0.

$$a_0 \, |0011\rangle = + \, |0010\rangle$$

03 **Creation**

Orbital 1 supplies one parity minus sign.

$$a_2^\dagger a_0 \, |0011\rangle = - \, |0110\rangle$$

Two orbitals show the eigenvalue problem explicitly

$$H_{N=1} = \begin{pmatrix} \epsilon_0 & t \\ t & \epsilon_1 \end{pmatrix}, \qquad E_- = \frac{\epsilon_0 + \epsilon_1}{2} - \sqrt{\left(\frac{\epsilon_0 - \epsilon_1}{2}\right)^2 + t^2}$$

01 The one-electron basis is {|01⟩, |10⟩}.

02 Hopping mixes occupation configurations.

03 SKQD will select a coordinate subspace, then diagonalize H there.

Derive hopping from fermions to qubits

01 **Orbital Hamiltonian**

The last term moves an electron.

$$H = \epsilon_0 n_0 + \epsilon_1 n_1 + t(a_0^\dagger a_1 + a_1^\dagger a_0)$$

02 **Pauli identity**

Both XX and YY preserve particle number together.

$$a_0^\dagger a_1 + a_1^\dagger a_0 = \tfrac{1}{2}(X_0 X_1 + Y_0 Y_1)$$

03 **One-electron block**

In basis |01⟩, |10⟩: now diagonalize 2×2.

$$H_{N=1} = \begin{pmatrix} \epsilon_0 & t \\ t & \epsilon_1 \end{pmatrix}$$

Now SKQD can begin



01 The quantum sampler discovers candidate basis labels.

02 The final eigenvalue comes from the target Hamiltonian projected into that basis.

The target is a specified eigenvalue problem

$$H_{\mathrm{tar}}|E_j\rangle = E_j|E_j\rangle, \qquad E_0 = \min_{\psi\neq 0} \frac{\langle\psi|H_{\mathrm{tar}}|\psi\rangle}{\langle\psi|\psi\rangle}$$

01 Geometry, basis, orbitals, electrons and sector define the target.

02 Agreement with CASCI is accuracy for that active-space model.

03 Carry nuclear and frozen-core energy offsets consistently.

Three dimensions—not one

Object	Dimension	Meaning
Active register	2^n	All qubit coordinates before extra constraints
Fixed spin sector	C(m,Nα) C(m,Nβ)	Occupation determinants in m spatial orbitals
Coherent Krylov space	rank(Φ) ≤ K	Span of K time-evolved vectors
Sampled coordinate space	D	Distinct selected basis labels

Six sampling times do not imply a six-dimensional final eigensystem.

Count determinants before interpreting a result

$$D_{N_\alpha,N_\beta} = \binom{m}{N_\alpha}\binom{m}{N_\beta}$$

01 CAS(8e,6o), balanced spin: 15 × 15 = 225.

02 CAS(10e,6o), balanced spin: 6 × 6 = 36.

03 CAS(6e,6o), balanced spin: 20 × 20 = 400.

Checkpoint: count the CAS(8e,6o) spaces

01 **All occupations**

12 spin orbitals, any electron number.

$$D_{\mathrm{Fock}} = 2^{12} = 4096$$

02 **Fixed electron count**

Eight electrons, spin unrestricted.

$$D_{N=8} = \binom{12}{8} = 495$$

03 **Four up + four down**

Fixed-spin sector of six spatial orbitals.

$$D_{4\alpha,4\beta} = \binom{6}{4}^2 = 225$$

Occupation bits encode antisymmetric states

$$n_p = a_p^\dagger a_p \longleftrightarrow \frac{I - Z_p}{2}, \qquad a_p|b\rangle = (-1)^{\sum_{q<p} b_q} b_p |b \setminus p\rangle$$

01 Before tapering, a Jordan–Wigner bitstring labels an occupation determinant.

02 The fermionic sign depends on the orbital ordering.

03 After a general tapering Clifford, the interpretation can change.

Model truncation versus exact representation change

Active-space choice

Select active orbitals and active electrons.

Freeze or discard other degrees of freedom.

Changes the physical approximation being solved.

Symmetry tapering

Choose a valid symmetry sector.

Use an exact isometry to a smaller register.

Preserves that sector’s spectrum—not every sector at once.

Classical Krylov: a small Hamiltonian-generated space

$$\mathcal{K}_K(H, r) = \mathrm{span}\{|r\rangle, H|r\rangle, \ldots, H^{K-1}|r\rangle\}$$

01 Do not form the whole spectrum if a small subspace is sufficient.

02 An arbitrary vector in this space is $p_{k-1}(H)|r\rangle$.

03 The initial reference determines which eigenspaces are accessible.

Why Krylov helps: polynomial spectral filtering

$$|r\rangle = \sum_j \gamma_j |E_j\rangle \quad \Longrightarrow \quad p(H)|r\rangle = \sum_j \gamma_j p(E_j)|E_j\rangle$$

01 Choose a polynomial that favors the low-energy spectral components.

02 Rayleigh–Ritz finds the best energy inside the available span.

03 Convergence depends on overlap and spectral separation.

Build the polynomial filter one step at a time

01 **Eigen-expansion**

The reference must overlap the desired eigenstate.

$$|r\rangle = \sum_j \gamma_j |E_j\rangle$$

02 **Apply a polynomial**

Powers of H reweight each eigencomponent.

$$p(H)|r\rangle = \sum_j \gamma_j p(E_j)|E_j\rangle$$

03 **Target low energy**

The subspace optimisation seeks such a filter.

$$|p(E_0)| \gg |p(E_j)| \quad (j > 0)$$

Derive Rayleigh–Ritz in a nonorthogonal basis

$$|\psi\rangle = \Phi c, \quad H_{\mathcal{K}} = \Phi^\dagger H \Phi, \quad S = \Phi^\dagger \Phi, \qquad E(c) = \frac{c^\dagger H_{\mathcal{K}} c}{c^\dagger S c}$$

01 Φ contains the basis vectors as columns.

02 The Gram matrix S measures their geometry.

03 A constrained stationary point gives H𝒦c = ESc.

A Ritz solution still has an external residual

01 **Trial state**

The projection is exact only within the chosen span.

$$|\psi_D\rangle = \Phi c, \quad \Phi^\dagger H \Phi c = E \Phi^\dagger \Phi c$$

02 **Residual**

It can point outside the trial subspace.

$$|r_{\mathrm{ext}}\rangle = H\,|\psi_D\rangle - E\,|\psi_D\rangle$$

03 **Galerkin orthogonality**

Zero projected residual is not zero full residual.

$$\Phi^\dagger\,|r_{\mathrm{ext}}\rangle = 0$$

The overlap matrix is geometry, not decoration

$$S_{jk} = \langle \phi_j | \phi_k \rangle, \qquad c^\dagger S c = \|\Phi c\|^2, \qquad S \succeq 0$$

01 Distinct vectors need not be linearly independent.

02 Nearly dependent vectors produce tiny Gram eigenvalues.

03 Orthonormal coordinates give S = I.

Lanczos constructs a stable Hermitian projection

$$w = Hq_j - \beta_j q_{j-1}, \quad \alpha_j = q_j^\dagger w, \quad w \leftarrow w - \alpha_j q_j, \quad q_{j+1} = w/\|w\|$$

01 The exact-arithmetic projected matrix is tridiagonal.

02 Finite precision can require reorthogonalization.

03 Davidson and sparse Lanczos are classical solver options later too.

Two guarantees—and a limitation

$$E_{\mathcal{V}} \geq E_0, \qquad \mathcal{V} \subseteq \mathcal{W} \Rightarrow E_{\mathcal{W}} \leq E_{\mathcal{V}}, \qquad r_H = H|\psi\rangle - E|\psi\rangle$$

01 Exact projection is variational for the same target sector.

02 Nested spaces cannot worsen the optimal energy.

03 A small residual identifies an approximate eigenstate—not automatically the ground state.

A four-state chain shows why expansion works

Selected basis	Dimension	Ritz energy	Error vs full exact
{|0⟩, |1⟩}	2	−0.0605551	0.00536694
Add |2⟩	3	−0.0658724	0.00004970
Add |3⟩	4	−0.0659221	~0

Nested, exact projected spaces cannot increase the minimum Ritz energy.

Quantum Krylov replaces powers by real-time circuits

$$|\phi_k\rangle = U^k|r\rangle, \qquad U = e^{-iH_{\mathrm{tar}}\Delta t}, \qquad \mathcal{K}_K(U, r)$$

01 This is exactly a polynomial Krylov space of U.

02 It is not generally the same finite-order space as $\mathscr{K}_{\mathrm{k}}$(H,r).

03 Different times generate phase patterns that a linear combination can exploit.

Real time by itself does not lower the energy

$$\langle r|e^{iHt}He^{-iHt}|r\rangle = \langle r|H|r\rangle$$

01 Each exact time-evolved basis state has the reference energy.

02 The improved energy comes from the final optimized linear combination.

03 QKS is subspace spectral extraction, not physical cooling.

Real-time QKS: calculate both matrices

01 **State preparation**

Start from $|+\rangle$, evolve for time δ.

$$H = Z, \quad |\phi_1\rangle = \frac{1}{\sqrt{2}}(e^{-i\delta}|0\rangle + e^{i\delta}|1\rangle)$$

02 **Overlap matrix**

The two states may be nearly parallel.

$$S = \begin{pmatrix} 1 & \cos\delta \\ \cos\delta & 1 \end{pmatrix}$$

03 **Projected target**

Complex off-diagonal matrix elements matter.

$$H_K = \begin{pmatrix} 0 & -i\sin\delta \\ i\sin\delta & 0 \end{pmatrix}$$

A complete one-qubit example

$$H = Z, \quad |r\rangle = |+\rangle, \quad |\phi_1\rangle = \frac{e^{-it}|0\rangle + e^{it}|1\rangle}{\sqrt{2}}$$

01 The target eigenvalues are +1 and −1.

02 Both prepared basis states have energy zero.

03 Two independent time states can span the entire qubit space.

Calculate both projected matrices

$$S = \begin{pmatrix} 1 & \cos t \\ \cos t & 1 \end{pmatrix}, \qquad H_{\mathcal{K}} = \begin{pmatrix} 0 & -i \sin t \\ i \sin t & 0 \end{pmatrix}$$

01 The off-diagonal Hamiltonian element is imaginary.

02 Discarding its imaginary part destroys the information.

03 $\det(H_{\mathcal{K}} - ES) = \sin^2(t)(E^2 - 1)$.

The ground state is a linear combination

$$|\phi_1\rangle - e^{-it}|\phi_0\rangle = \sqrt{2}\, i \sin t\, |1\rangle$$

01 The |0⟩ amplitude cancels exactly.

02 Small t makes the cancellation numerically delicate.

03 Knowing the coefficients is not the same as preparing the combination on hardware.

Try it: reconstruct the exact ground state

01 **Good time choice**

Two orthogonal Krylov basis states.

$$\delta = \pi/2 \quad \Longrightarrow \quad S = I$$

02 **Ritz coefficients**

Relative complex phase fixes interference.

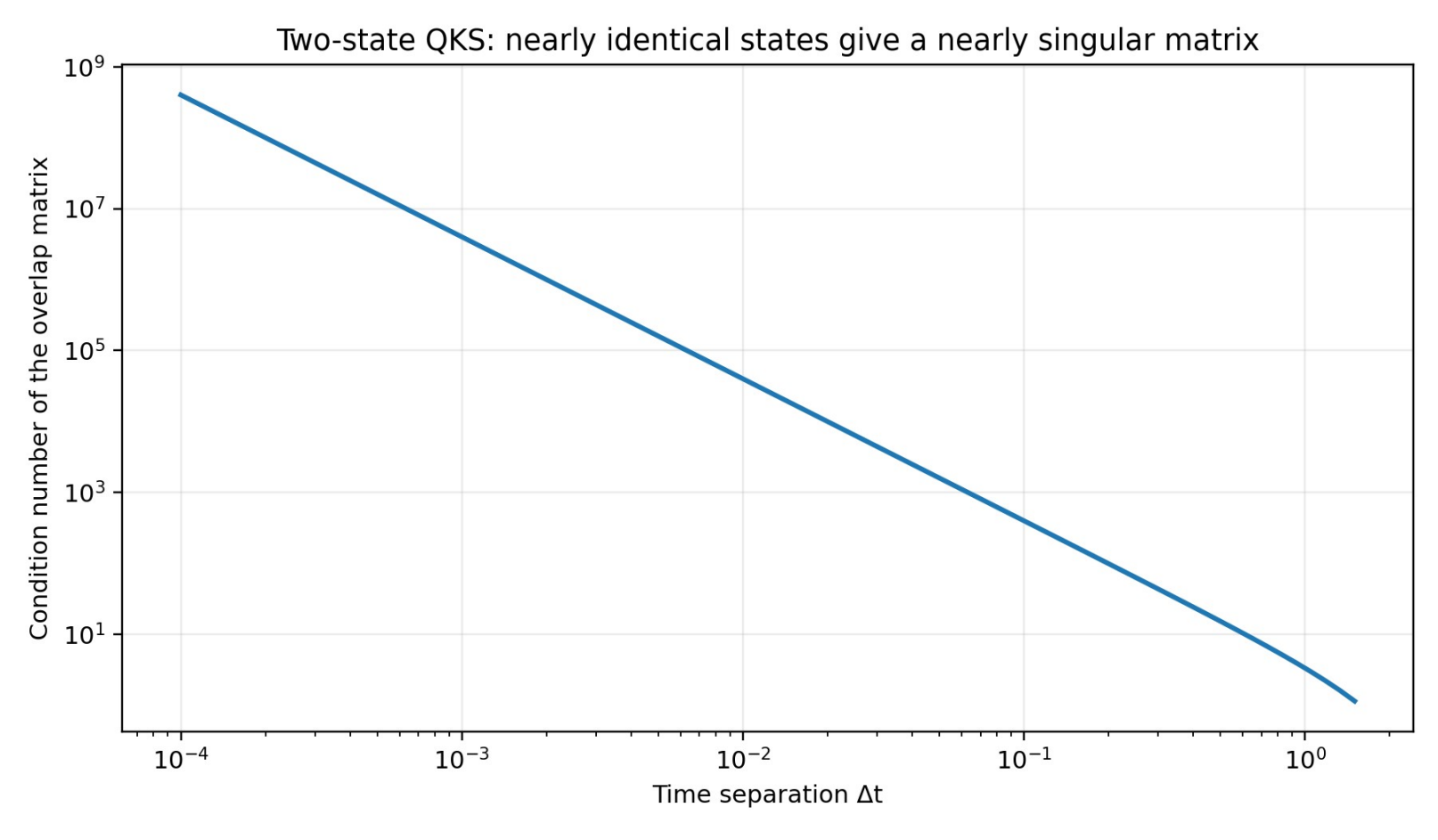
$$c = \frac{1}{\sqrt{2}}(1,\, -i)^T$$

03 **Physical state**

The eigenstate of Z with energy −1.

$$\Phi c = \frac{|+\rangle - |-\rangle}{\sqrt{2}} = |1\rangle$$

When a correct subspace becomes numerically fragile



$\kappa(S) = (1 + |\cos \Delta t|) / (1 - |\cos \Delta t|)$.

For small Δt, $\kappa(S) \approx 4/\Delta t^2$.

Useful exact span ≠ robust measured eigensystem.

Aliasing: different energies can have the same phase

$$e^{-iE_j\Delta t} = e^{-iE_\ell\Delta t} \quad \Longleftrightarrow \quad (E_j - E_\ell)\Delta t = 2\pi m$$

01 At aliased phases, time samples cannot distinguish those components.

02 Tiny times create near dependence; special large times can create exact dependence.

03 Choose the time grid with spectral scale and implementation cost in view.

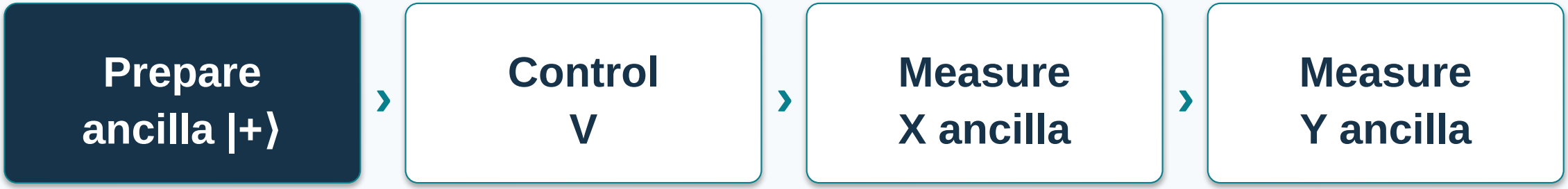
What a Hadamard test measures

Prepare $|+\rangle$ ancilla › **Controlled V on $|r\rangle$** › **Measure X or Y ancilla**

01 $\langle X_a \rangle = \mathrm{Re}\langle r|V|r\rangle$; $\langle Y_a \rangle = \mathrm{Im}\langle r|V|r\rangle$.

02 For transition terms, $V = U_j^\dagger P_\alpha U_k$; then sum $h_\alpha \langle V \rangle$.

Hadamard test: phase information is the product



01 Re⟨r|V|r⟩ and Im⟨r|V|r⟩ reconstruct the complex overlap.

02 A final all-zero probability alone gives only $|\langle r|V|r\rangle|^2$.

Equal time spacing can reduce distinct overlap data

$$S_{jk} = s(t_k - t_j), \qquad (H_{\mathcal{K}})_{jk} = h(t_k - t_j)$$

01 For exact evolution with the same target H, the matrices depend on time lags.

02 Repeated-unitary overlap data remain lag-structured.

03 The projected H need not be lag-structured when the repeated approximation does not commute with H.

Time evolution has its own approximation budget

$$e^{-it\sum_\alpha h_\alpha P_\alpha} \approx \left(\prod_\alpha e^{-i(t/r)h_\alpha P_\alpha}\right)^r$$

01 Record product-formula order, term order and number of steps.

02 Diagonal does not mean free: a long Z string may still need entanglers.

03 Compare compiled native gates and depth—not symbolic term counts alone.

Stable QKS: whiten a retained positive Gram subspace

$$S = V\Lambda V^\dagger, \quad W = V_R \Lambda_R^{-1/2}, \quad W^\dagger S W = I, \quad (W^\dagger H_{\mathcal{K}} W)y = Ey$$

01 Hermitize estimates; inspect the Gram spectrum.

02 Keep eigenvalues above a declared, noise-aware cutoff.

03 Recover c = Wy and check the residual.

Canonical whitening: the full algorithm

01 **Decompose the Gram**

Negative/noisy tiny modes cannot set valid norms.

$$\widehat{S} = V \operatorname{diag}(\lambda_1, \ldots, \lambda_K) V^\dagger$$

02 **Retain and whiten**

Keep only stable, positive eigenmodes.

$$W = V_R \operatorname{diag}(\lambda_i^{-1/2})_{i \in R}$$

03 **Solve Hermitian Ritz**

Do not explicitly form $S^{-1}H$.

$$W^\dagger \widehat{H} W \, y = E y, \qquad c = W y$$

Where measurement noise enters the eigenvalue

$$\delta E \approx c^\dagger(\delta H_{\mathcal{K}} - E\,\delta S)c, \qquad c^\dagger S c = 1$$

01 Small Gram eigenvalues can require large Euclidean coefficient norms.

02 Noise in S and H$\mathscr{K}$ can then be amplified.

03 Use repeated estimates, cutoff sweeps and retained-rank diagnostics.

Before leaving coherent QKS

What the QPU supplies

A family of preparable basis states.

Complex overlaps and Hamiltonian transition data.

Circuits repeated at enough shots for the required precision.

What the classical solver supplies

A rank-regularized projected problem.

Ritz coefficients and energy estimates.

A diagnosis of conditioning—not automatic state preparation.

SKQD changes the final basis

Coherent QKS

Final basis: $|\varphi_0\rangle, \ldots, |\varphi_{k-1}\rangle$.

Needs transition matrix elements.

Solves a generally nonorthogonal projected problem.

SKQD

Final basis: distinct selected coordinate states.

Quantum states propose labels by sampling.

Builds target matrix elements classically and diagonalises them.

Two spaces, two different jobs

Time-evolved quantum states › **Computational basis samples** › **Selected label set $\mathscr{D}$** › **Classical target projection**

01 The coherent span generates the sample distributions.

02 The coordinate span is the space in which the final eigenproblem is solved.

Why S = I—and what it does not mean

$$B_{\mathcal{D}} = [|b_1\rangle \ \cdots \ |b_D\rangle], \qquad B_{\mathcal{D}}^{\dagger} B_{\mathcal{D}} = I_D, \qquad H_{\mathcal{D}} = B_{\mathcal{D}}^{\dagger} H_{\mathrm{tar}} B_{\mathcal{D}}$$

01 Deduplicate the sampled labels before forming the basis.

02 Counts influence selection; they are not the overlap matrix.

03 S = I removes Gram conditioning, not subspace incompleteness.

Sampling loses phase; the target Hamiltonian supplies couplings

$$p_k(b) = |\langle b|\psi(t_k)\rangle|^2, \qquad H_{\mathcal{D}}c = Ec$$

01 Samples do not reconstruct the full quantum wavefunction.

02 Known signed or complex matrix elements determine the Ritz coefficients.

03 The output is not the average of diagonal energies of sampled strings.

From Pauli strings to selected matrix elements

01 **Selected basis**

Each distinct computational-basis label is orthonormal.

$$B_D = [|b_1\rangle \ \cdots \ |b_D\rangle]$$

02 **Full target block**

Use the correct tapered target, not the cheap driver.

$$H_D = B_D^\dagger H_{\text{target}} B_D$$

03 **Pauli action**

Bit flips and phases build matrix entries efficiently.

$$X|0\rangle = |1\rangle, \quad Y|0\rangle = i|1\rangle, \quad Z|1\rangle = -|1\rangle$$

A useful surprise: coordinate selection can leave the coherent span

$$H = X, \quad |r\rangle = |+\rangle, \quad e^{-itH}|r\rangle = e^{-it}|+\rangle$$

01 Coherent QKS stays one-dimensional at energy +1.

02 Sampling |+⟩ can discover both |0⟩ and |1⟩.

03 Their coordinate span contains the ground state |−⟩ at energy −1.

A one-vector Krylov span can discover a better space

01 **Reference eigenstate**

Coherent time Krylov span remains 1D.

$$H = X, \quad |r\rangle = |+\rangle, \quad U(t)\,|r\rangle = e^{-it}\,|+\rangle$$

02 **Z-basis samples**

Both computational coordinates are observed.

$$\Pr(0) = \Pr(1) = \tfrac{1}{2}$$

03 **Selected diagonalisation**

The selected space contains $|-\rangle$ though QKS span does not.

$$\operatorname{span}\{|0\rangle, |1\rangle\} = \mathbb{C}^2, \quad E_D = -1$$

Separate the target from the sampling driver

$$|\psi_k\rangle = e^{-iH_{\mathrm{drive}}t_k}|r\rangle, \qquad (H_{\mathcal{D}})_{ab} = \langle b_a|H_{\mathrm{tar}}|b_b\rangle$$

01 The cheap driver generates a proposal distribution.

02 The full target determines the energy in the selected space.

03 A driver need not accurately reproduce all target dynamics to be a useful proposal.

An identity shift only adds an overall phase

01 **Driver shift**

Identity adds no transitions.

$$H'_{\mathrm{drive}} = H_{\mathrm{drive}} + cI$$

02 **Unitary evolution**

The extra factor is a global phase.

$$e^{-iH'_{\mathrm{drive}}t} = e^{-ict}e^{-iH_{\mathrm{drive}}t}$$

03 **Sample histogram**

Sampling statistics are unchanged.

$$|\langle b| e^{-iH'_{\mathrm{drive}}t} |r\rangle|^2 = |\langle b| e^{-iH_{\mathrm{drive}}t} |r\rangle|^2$$

Build a selected Pauli block without a full 2^n matrix

$$P|b\rangle = \omega_P(b)|b \oplus x_P\rangle, \qquad \langle a|P|b\rangle = \omega_P(b)\,\delta_{a,b\oplus x_P}$$

01 X and Y determine the bit-flip mask x_p.

02 Y and Z determine a sign or complex phase.

03 Use a hash table of selected labels to assemble only retained connections.

Noisy sampling can still give a variational selected energy

$$E_{\mathcal{D}} = \lambda_{\min}\left(B_{\mathcal{D}}^{\dagger} H_{\mathrm{tar}} B_{\mathcal{D}}\right) \geq E_0$$

01 Noise changes which labels are proposed.

02 Exact classical projection remains variational in the same valid sector.

03 Wrong-sector labels, inconsistent maps or noisy matrix elements can invalidate that comparison.

Captured ground-state weight bounds the energy error

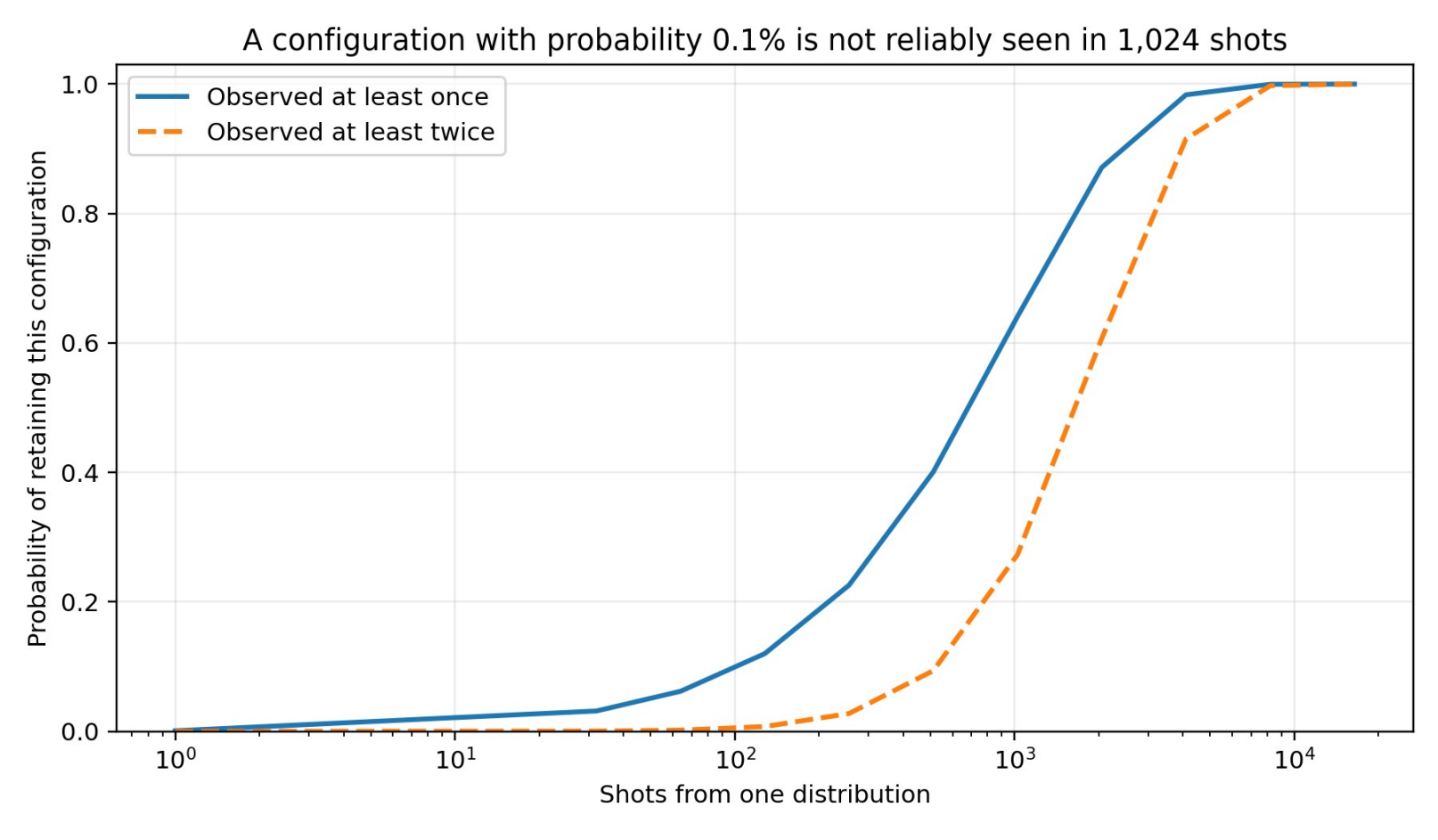
$$w = \langle E_0 | P_{\mathcal{D}} | E_0 \rangle > 0 \quad \Longrightarrow \quad 0 \leq E_{\mathcal{D}} - E_0 \leq (E_{\max} - E_0)(1 - w)$$

01 Normalize the projected ground state to obtain a trial vector.

02 Its ground-state fidelity is w.

03 The bound explains support selection; w is usually unknown in a real experiment.

A probability of 0.1% is not reliably seen in 1,024 shots



At least one count: about 64.1%.

At least two counts: about 27.3%.

These are probabilities for one fixed distribution.

Why a two-count threshold misses real configurations

01 **Model**

Mean count is 1.024.

$$C \sim \text{Binomial}(N, p), \quad N = 1024,\ p = 10^{-3}$$

02 **At least once**

About 36% of experiments never see it.

$$\Pr(C \geq 1) = 1 - (1 - p)^N \approx 0.6410$$

03 **At least twice**

Most experiments fail a count ≥2 rule.

$$\Pr(C \geq 2) = 1 - (1 - p)^N - Np(1 - p)^{N-1} \approx 0.2731$$

Multiple times: detection depends on the whole allocation

$$\Pr[b \text{ unseen}] = \prod_k (1 - p_k(b))^{N_k} \leq \exp\left[-\sum_k N_k p_k(b)\right]$$

01 More shots at an uninformative time may discover little.

02 Different times can make different configurations visible.

03 Thresholded retention is stricter than merely observing a label.

Two probability distributions you must not confuse

Ground-state weights

$|\langle b|E_0\rangle|^2$ measures the target state's support.

Useful for diagnosing captured weight in small reference calculations.

Not the distribution sampled by an arbitrary driver.

Driver sampling probabilities

$p_k(b) = |\langle b|\exp(-iH_{\text{drive}}\, t_k)|r\rangle|^2$.

Determine finite-shot discovery at each time.

Depend on the driver, reference, grid and noise.

Short-time sampling exposes the driver graph

01 **Drive distribution**

These probabilities determine the observed samples.

$$p_t(b) = |\langle b| e^{-iH_{\text{drive}}t} |r\rangle|^2$$

02 **Short-time transition**

For b≠r, direct transitions begin at order t.

$$\langle b| e^{-iH_{\text{drive}}t} |r\rangle = -it(H_{\text{drive}})_{br} + O(t^2)$$

03 **Disconnection**

Infinite shots cannot sample an unreachable label.

$$(H^k_{\text{drive}})_{br} = 0 \ \forall k \implies p_t(b) = 0$$

Sparsity is an assumption, not a universal chemistry fact

01 A small useful coordinate set makes sample-based diagonalisation practical.

02 Useful sparsity depends on the orbital basis and symmetry representation.

03 Strong correlation can increase the number of configurations that matter.

04 Sampling and classical expansion may both become bottlenecks.

Ankush’s four-reduction architecture

Exact Z_2 tapering › **Perturbative term screening** › **Compact driver + SKQD** › **Two-tier filtering**

01 Then expand the selected core using target-Hamiltonian connectivity.

02 Construct the full-target block and diagonalise classically.

Reduction 1: find commuting Pauli symmetries

$$v_P = (x_P|z_P), \quad \Omega = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \qquad G\Omega v_\tau^T = 0 \pmod 2$$

01 Represent Pauli commutation as binary linear algebra.

02 Select independent, mutually commuting symmetry generators.

03 Choose sector signs and record the exact embedding.

Work the binary symplectic commutation rule

01 **Binary Pauli labels**

Bit 1 records an X and/or Z component.

$$P \sim (x \mid z) \in \mathbb{F}_2^{2n}$$

02 **Check two Pauli terms**

They anticommute on each qubit: two minus signs.

$$P = X_0 Z_1, \qquad Q = Z_0 X_1$$

03 **Symplectic parity**

Thus P and Q commute.

$$x_P \cdot z_Q + z_P \cdot x_Q = 1 + 1 = 0 \pmod 2$$

A tapering example you can verify by hand

$$H = aZ_0 + bZ_1 + cX_0X_1, \quad \tau = Z_0Z_1, \quad H_{\tau=+1} = (a+b)Z + cX$$

01 The even-parity basis is $|00\rangle$, $|11\rangle$.

02 Z_0 and Z_1 act identically inside that sector.

03 X_0X_1 flips the two retained coordinates.

Do not confuse a parity sector with particle number

Sector reduction

Two commuting sectors of Z_0Z_1: +1 and −1.

Even sector is {|00⟩, |11⟩}.

Its effective target is gX + 2hZ + JI.

Physical interpretation

Choosing wrong sector gives wrong target energy.

A Clifford-tapered bit can encode a superposition.

Parity alone does not fix exact electron count.

After tapering, a bitstring need not be a determinant

$$J^\dagger J = I, \qquad H_{\mathrm{tar}} = J^\dagger H_{\mathrm{full}} J, \qquad |b\rangle \mapsto J|b\rangle$$

01 The lifted states remain orthonormal.

02 They may be symmetry-adapted combinations of occupation determinants.

03 S = I survives; a direct occupation interpretation may not.

Reduction 2: the manuscript’s EN term-screening score

$$\Delta\epsilon_k^{(2)} \approx -\frac{|\langle\Phi_0|c_k P_k|\Phi_k\rangle|^2}{E_k - E_{\mathrm{ref}}}, \qquad \eta_{\mathrm{prune}} = 10^{-7}\ \mathrm{Ha}$$

01 Preserve all diagonal I/Z-type terms in the reported policy.

02 Screen weak off-diagonal candidates by a second-order estimate.

03 Treat the quoted <0.1 mHa effect as reported behavior, not a universal bound.

Why individual Pauli scores need care

$$\left|\sum_k c_k \langle D|P_k|\Phi_0\rangle\right|^2 \neq \sum_k |c_k \langle D|P_k|\Phi_0\rangle|^2$$

01 Different Pauli terms can reach the same configuration and interfere.

02 Near-degenerate denominators invalidate a simple weak-coupling picture.

03 Pruning individual strings can break cancellations that preserve particle number.

Individual Pauli terms can interfere

01 **Same transition**

Different Pauli terms may connect the same two labels.

$$H_{br} = \sum_{\ell : r \to b} h_\ell \omega_\ell(r)$$

02 **Correct second order**

Add transition amplitudes before squaring.

$$\Delta E_{br}^{(2)} \propto \left| \sum_{\ell} h_\ell \omega_\ell \right|^2$$

03 **Cross terms**

Reinforcement and cancellation alter the score.

$$|a+b|^2 = |a|^2 + |b|^2 + 2\mathrm{Re}(a^* b)$$

Reduction 3: a cheap driver for support discovery

$$H_{\mathrm{drive}} = H_{\mathrm{diag}} + \sum_{j=1}^{L_{\mathrm{drive}}} c_j P_j, \qquad t_k \in \{0, .15, .30, .45, .60, .75\}$$

01 The manuscript ranks compact-drive candidates by $|c_j|$.

02 The general text says 10–30 off-diagonal terms.

03 Table 3 lists 80 for N_2 and 50 for CH_4: obtain molecule-specific settings.

Connected expansion: add target-coupled coordinates

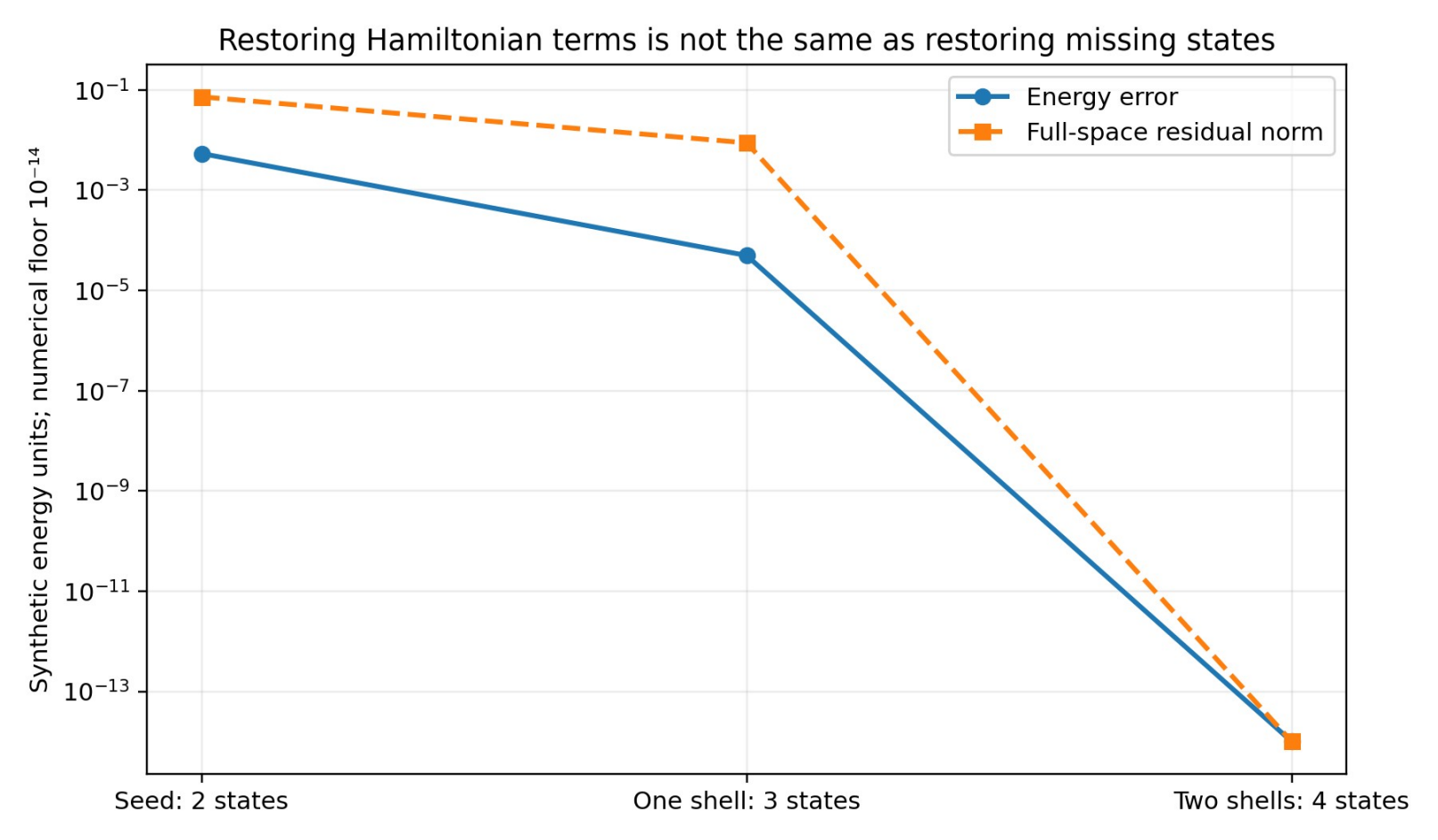
$$\mathcal{D}_{\mathrm{next}} = \mathcal{D}_{\mathrm{core}} \cup \{b : \exists a \in \mathcal{D}_{\mathrm{core}},\ |H_{ab}| \geq \epsilon_{\mathrm{expand}}\}$$

01 The displayed manuscript rule is a core-connected expansion.

02 One shell is different from repeated closure to all reachable states.

03 Record the threshold, number of shells and final dimension.

Restoring terms is not restoring missing states



A two-state seed misses part of the chain ground state.

One shell leaves a small nonzero error and residual.

Two shells recover the complete four-state target.

Reduction 4: two distinct filters

$$C_{\max}(b) \geq 2, \quad f_{\max}(b) \geq 0.001; \qquad I_b = \frac{|\langle \Phi_0 | H | b \rangle|^2}{|E_{\mathrm{ref}} - H_{bb}|} \geq 10^{-6}\ \mathrm{Ha}$$

01 Tier 1: repeated count and frequency.

02 Tier 2: energetic coupling to the reference.

03 Edge policies matter: protect the reference; handle zero denominators explicitly.

Water’s two Tier-1 thresholds are one count rule

01 **Counts**

At least two observations at one time point.

$$C_{\max}(b) \geq N_{\min} = 2$$

02 **Frequency**

At least 0.1% observed frequency.

$$C_{\max}(b)/1024 \geq 0.001$$

03 **Combined integer rule**

At 1024 shots they are the same threshold.

$$C_{\max}(b) \geq \max\{2, \lceil 1.024 \rceil\} = 2$$

A physical configuration can have zero reference score

$$H_{02} = 0, \qquad H_{01} \neq 0, \quad H_{12} \neq 0 \quad \Longrightarrow \quad 0 \longleftrightarrow 1 \longleftrightarrow 2$$

01 Configuration 2 is reached indirectly through configuration 1.

02 Its Hartree–Fock-reference screening score is zero.

03 A multireference residual can identify directions missed by a reference-only filter.

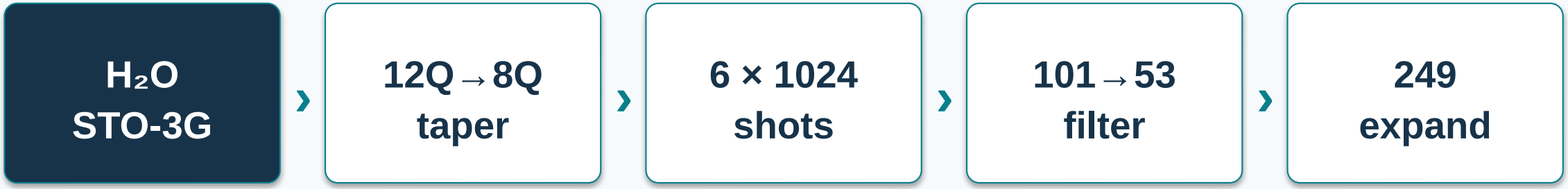
The reported water experiment

12 spin orbitals › **8 tapered qubits** › **6 × 1,024 shots** › **53 filtered labels** › **249 expanded basis objects**

01 H_2O: R = 1.0 Å, angle 104.5°, STO-3G; Table 3 gives CAS(8e,6o).

02 The manuscript reports physical Quantinuum H2-1 execution.

Trace the exact water workflow in order



01 Full target projected into the expanded selected space, then classically diagonalized.

02 Raw and filtered branches differ; 249 is the reported filtered expanded dimension.

Reported: [M] §§4.1–4.2, pp. 9–10

Water: preserve the reported comparison

Method	Qubits	ZZPhase	Depth	Error (mHa)
Unpruned "standard QKS"	12	5,436	18,572	>50
Classical statevector SKQD	8	150	428	0.1836
1,024-shot emulation	8	150	428	0.1836
Physical H2-1, raw	8	150	428	0.1113
Physical H2-1, filtered	8	150	428	0.3095

Raw support has a lower reported error than filtered support; filtering trades accuracy against support size.

[M] Table 1, p. 10 • Not independently reproduced

What does a 249-dimensional water space represent?

$$\frac{249}{2^8} = 97.27\%, \qquad \binom{6}{4}\binom{6}{4} = 225$$

01 249 is almost the full 256-dimensional tapered coordinate space.

02 225 is the balanced-spin occupation count for CAS(8e,6o).

03 Clarify the basis map and particle/spin sector before interpreting "249 determinants."

An interpretation check: 249 is bigger than 225

01 **Fixed-spin CAS**

CAS(8e,6o) with four alpha and four beta electrons.

$$\binom{6}{4}\binom{6}{4} = 225$$

02 **8-qubit register**

Reported 249 nearly fills the tapered coordinate space.

$$2^8 = 256, \qquad 249/256 \approx 97.27\%$$

03 **Question, not verdict**

Which exact symmetry/particle sector did the expansion retain?

$$225 < 249 < 256$$

The broader reported active-space suite

Molecule	Active space	Tapered q	Drive terms	Max error, mHa
H_2O	CAS(8e,6o)	8	10–30	0.3095
F_2	CAS(10e,6o)	7	30	0.0000
N_2	CAS(6e,6o)	7	80	0.2854
CH_4	CAS(6e,6o)	8	50	0.0016
C_2H_4	CAS(2e,2o)	4	10	0.0000
C_6H_6	CAS(6e,6o)	7–8	20	0.0012

Do not label every geometry as independently validated physical hardware.

[M] Table 3, p. 11 • Reported values

Reported 0.0000 mHa: interpret the precision

What the source reports

F_2 result: 0.0000 mHa against CASCI.

Several small active-space curves show 0.0000 mHa at tabulated precision.

Chemistry-threshold pass rates use the source reference.

What it does not establish

Agreement to displayed decimals is not mathematical exactness.

CASCI reference is not all-electron full-basis FCI.

Novelty or quantum advantage needs stronger controls.

What the results support—and what they do not establish

Supported as manuscript reports

A concrete integration of tapering, compact sampling and classical projection.

Low active-space errors in the listed benchmarks.

Large compiled-resource reductions for the compared workflows.

Not established by these tables alone

Quantum advantage over strong same-target classical methods.

Universal protection from noise or missing support.

Full-molecule accuracy or independently verified hardware provenance.

A replication needs these missing inputs

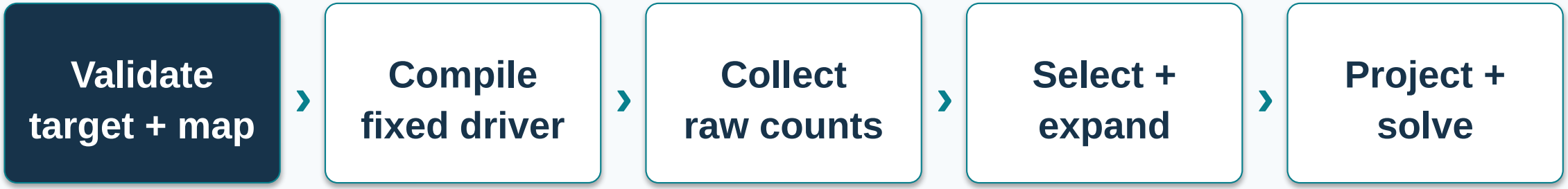
01 Molecular integrals, orbital ordering, active-space definition and energy offsets.

02 Symmetry generators, sector signs, embedding and tapered reference.

03 Actual drive list, time units, product formula and compiled circuits.

04 Raw counts, all filter edge policies, expansion rules and precise reference energies.

An auditable end-to-end implementation



01 Hash and version each target, driver and selected label set.

02 Save the raw data so every threshold choice can be replayed.

Implementation contract: no invisible choices

At each stage	Must be saved	Why it matters
Molecule + target	Geometry, integrals, offsets, CAS	Defines the physical eigenproblem
Symmetry reduction	Generators, signs, isometry	Defines basis and valid sector
Driving circuits	Terms, t units, Trotter, compiler	Rebuilds physical preparation
Samples	Raw per-circuit counts, shot seeds	Replays finite-shot selection
Classical selection	Filter, expansion, eigensolver	Reproduces final basis/energy

A native gate count and one final energy are not enough for exact replication.

Five laboratory tasks to master the mechanics

Lab	Question	What is checked
A: one qubit	Does coherent QKS recover ±1?	Complex transitions, whitening, aliasing
B: four-state chain	What does connected expansion recover?	Variational energies and full residuals
C: repeated sampling	What does a finite budget miss?	Count filters, uniform control, fresh trials
D: Hubbard dimer	Are the fermionic signs correct?	Sector matrix and analytic ground energy
E: reusable functions	Can a selected block avoid full storage?	Bitmask Pauli assembly versus dense test

All manuscript hardware numbers remain reported data; all laboratory numbers have explicit toy-model inputs.

Numerical lab: verify the Hubbard-dimer sector

01 **Active sector**

Four determinant configurations.

$$\mathcal{H}_{N_\uparrow=N_\downarrow=1} = \mathrm{span}\{|0101\rangle, |0110\rangle, |1001\rangle, |1010\rangle\}$$

02 **Hamiltonian**

Fermionic hopping and on-site repulsion.

$$H_{\mathrm{Hub}} = -t\sum_\sigma (c_{0\sigma}^\dagger c_{1\sigma} + \mathrm{h.c.}) + U\sum_i n_{i\uparrow}n_{i\downarrow}$$

03 **Check exact number**

For U=4, t=1: $E_0=2-2\sqrt{2}\approx -0.828427$.

$$E_0 = \tfrac{1}{2}\left(U - \sqrt{U^2+16t^2}\right)$$

The Hubbard dimer checks the chemistry machinery

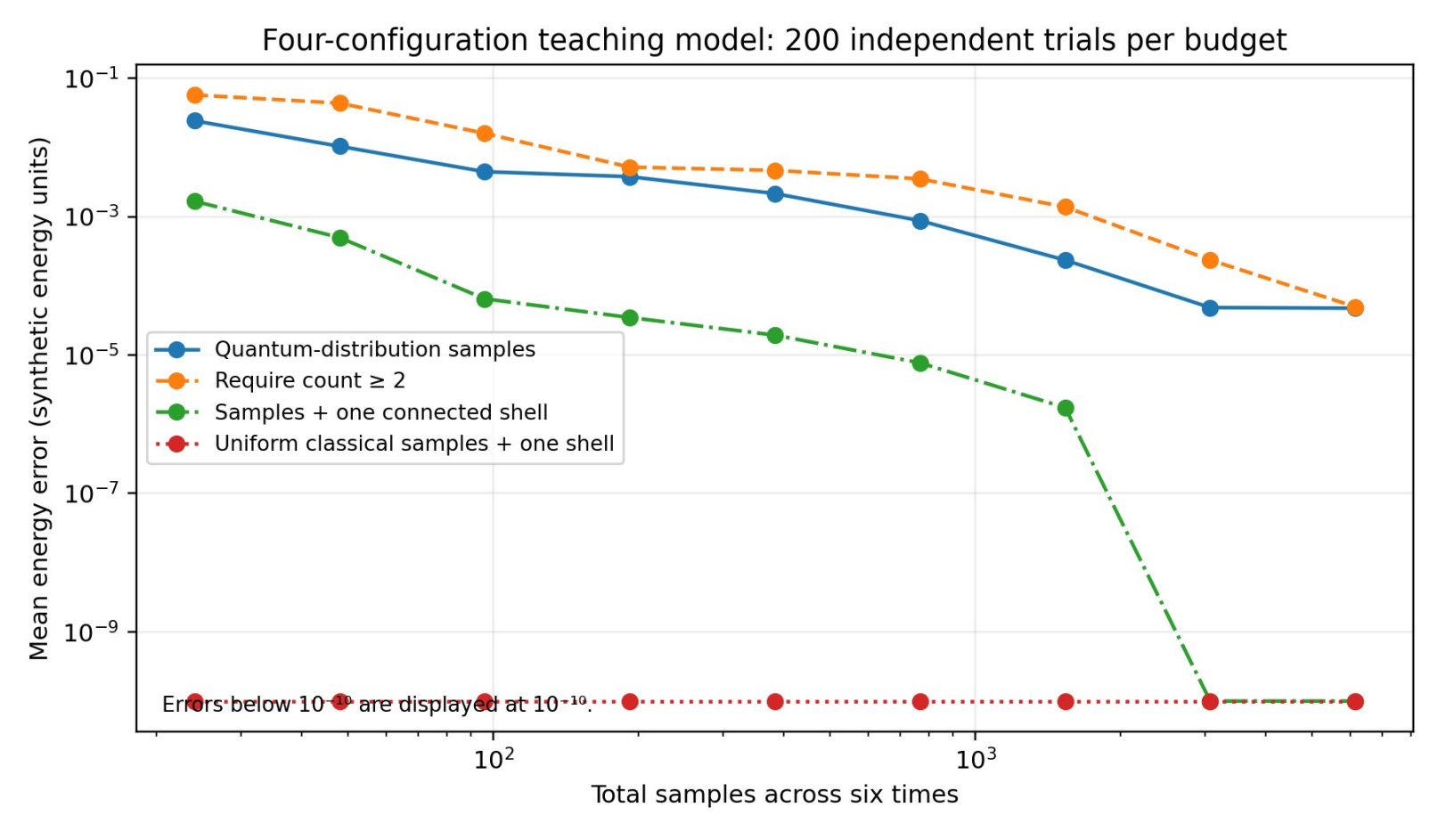
$$H = -t \sum_{\sigma} (a_{0\sigma}^{\dagger} a_{1\sigma} + \text{h.c.}) + U \sum_{j} n_{j\uparrow} n_{j\downarrow}, \qquad E_0 = \frac{U - \sqrt{U^2 + 16t^2}}{2}$$

01 With t = 1 and U = 4, $E_0 = 2 - \sqrt{8} = -0.828427124746\ldots$

02 Restrict to one electron of each spin.

03 Build the fermionic operators first; then project and compare.

Keep the classical baseline—even when it wins



200 independent trials per budget.

The quantum-distribution proposal uses six times.

Uniform classical sampling plus expansion is a control, not a hidden omission.

Three source clarifications to resolve early

01 The listed water gates sum to 824, while the manuscript states 833 total gates.

02 The expansion threshold and number of expansion shells are not fully specified.

03 The general 10–30 drive-term range differs from N_2 = 80 and CH_4 = 50 in Table 3.

QKS, SKQD and QITE solve different intermediate problems

	Coherent QKS	SKQD	Local-unitary QITE
Quantum role	Generate + characterize states	Propose basis labels	Update an evolving state
Classical core	Generalized eigensystem	Selected target eigensystem	Generator linear systems
Main bottleneck	Transition data + conditioning	Support + classical dimension	Measurements + update depth
Output by default	Energy and coefficients	Energy and coefficients	An evolved quantum state

Match the target and output task before matching a resource budget.

Count the full executed workload

$$N_{\mathrm{shots}}^{\mathrm{total}} = \sum_{c} N_c, \qquad G_{2q}^{\mathrm{executed}} = \sum_{c} N_c \, g_{2q}(c)$$

01 Report per-circuit depth and executed entangler volume separately.

02 Include calibration, discarded shots, repeated batches and tuning runs.

03 Include classical wall time, memory and final selected dimension.

Account for the total experimental budget

01 **Measure outcomes**

Report executions for every circuit and repetition.

$$N_{\text{total shots}} = \sum_{k=0}^{K-1} N_k$$

02 **Gate workload**

Per-shot compiled 2Q operations, not just per-circuit depth.

$$G_{2q,\text{executed}} = \sum_{k} N_k \, G_{2q,k}$$

03 **Measure the task**

Same target, same reference, same selection policy.

$$\varepsilon_E = \left| E_{\text{method}} - E_{\text{target}} \right|$$

Hardware credits: use the backend’s actual accounting

$$\mathrm{HQC} = 5 + \frac{N_{1q} + 10N_{2q} + 5N_m}{5000}\,C$$

01 This is an official representative formula, not a cash-price quote.

02 The fixed submission term is outside the shot-dependent term.

03 Use the actual compiled backend estimate and account terms.

Ablations that identify where the accuracy comes from

Controlled change	Keep fixed	Question
Quantum vs classical proposal	Target, final-size policy, expansion	Does the quantum sampler add useful support?
Raw vs two-tier filtering	Same exported count batch	What accuracy–dimension tradeoff is introduced?
No / one / repeated expansion	Same seed support	How much work is classical expansion doing?
Full vs compact driver	Budget, reference, target	What is lost through drive restriction?
Tapered QKS vs tapered SKQD	Same representation and compiler	What is caused by the final-basis change?

An end-to-end engineering comparison is useful; a mechanism claim requires controlled changes.

Is the quantum sampler doing unique work?

Quantum-seeded support

Run the specified driver and sample at fixed shot budget.

Filter and expand with the full target.

Measure final Ritz error and classical expansion cost.

Classical-only controls

Start from HF + connected determinant shells.

Uniform or weighted classical determinant proposals.

Match final dimension, post-processing and target.

Use diagnostics with compatible dimensions

$$r_{\text{out}} = (I - P_{\mathcal{D}})H_{\text{tar}}|\psi_{\mathcal{D}}\rangle, \qquad \Delta E = E_{\mathcal{D}_{\text{old}}} - E_{\mathcal{D}_{\text{new}}}$$

01 Do not subtract D×D and (D+ΔD)×(D+ΔD) matrices directly.

02 Track full residuals, energy changes, discovery saturation and solve tolerances.

03 Energy stabilization alone can mean the proposal is stuck.

Research direction 1: adaptive shot allocation

$$\Pr[b \text{ unseen}] \leq e^{-\sum_k N_k p_k(b)}, \qquad \sum_k N_k = N_{\text{budget}}$$

01 Use pilot samples to estimate useful novelty at different times.

02 Reserve some exploration shots; rare unseen states have no direct estimates.

03 Evaluate at a fixed total budget over independent runs.

Research direction 2: score external correlated residuals

$$r_a = \sum_{b \in \mathcal{D}} H_{ab} c_b, \qquad I_a^{\mathrm{corr}} \approx \frac{|r_a|^2}{|E - H_{aa}|}$$

01 Use the current correlated Ritz vector—not only Hartree–Fock.

02 Select external directions with meaningful residual coupling.

03 Handle near-degeneracy and denominator regularization explicitly.

From HF scores to correlated-residual scores

01 **Current Ritz vector**

Known after classical diagonalisation.

$$|\psi_D\rangle = \sum_{a \in D} c_a \, |a\rangle$$

02 **External coupling**

Can be nonzero even when H(br)=0.

$$r_b = \langle b| \, H_{\mathrm{target}} \, |\psi_D\rangle = \sum_{a \in D} H_{ba} c_a$$

03 **Selection heuristic**

Rank uncovered directions, with a denominator floor.

$$I_b = \frac{|r_b|^2}{\max(|H_{bb} - E_D|, \delta_{\mathrm{den}})}$$

Research direction 3: adapt the driver carefully

Potential benefit

Use missed or high-residual configurations to choose new excitations.

Compare coherent groupings against isolated Pauli ranking.

Seek more useful support per executed entangler.

New costs and risks

Feedback makes the quantum stage adaptive rather than one-shot.

More terms may improve reachability but deepen circuits.

Selection can overfit a small benchmark or violate conservation.

A QITE–SKQD hybrid needs a nonredundant purpose

Potentially meaningful task

Use QITE-produced states as a competing support proposal.

Prepare the selected correlated state for later quantum observables.

Compare state quality at a fixed preparation and measurement budget.

A redundant energy-only task

First build a small known $H_{\mathscr{D}}$.

Then apply QITE merely to estimate its smallest eigenvalue.

Classical diagonalisation already solves that small matrix.

What would a useful hybrid actually output?

Energy-only hybrid

SKQD yields an explicit small classical H_D.

Classical diagonalization already finds its lowest eigenvalue.

QITE on the same known small H_D may simply add cost.

Potentially nontrivial task

Improve the quantum sampling distribution before support is known.

Prepare a usable quantum eigenstate or evaluate a new observable.

Compare all new costs with classical solutions.

A research plan with clear decision points

Reproduce one target › **Replay raw counts** › **Run matched ablations** › **Choose one hypothesis** › **Scale with strong controls**

01 First establish what is already explained by classical expansion.

02 Then optimize the bottleneck that actually dominates the measured workload.

Explain these without looking at the formulas

01 Why can QKS improve the energy when each exact real-time basis state has the same energy?

02 Why does SKQD have S = I without reconstructing quantum amplitudes?

03 Why does a full-target selected block not recover every missing configuration?

04 What exactly would you need to reproduce Ankush’s 249-dimensional water calculation?

A seven-session study plan

Session	Focus	Concrete output
1	Chemistry + Rayleigh–Ritz	Derive H$\mathscr{K}$c = ESc and determinant counts
2	Coherent QKS + conditioning	Solve the Z example and run Lab A
3	Sampled spaces + probabilities	Derive S = I and binomial discovery
4	Four manuscript reductions	Draw every input and output
5	Reported molecular results	Complete the reproduction audit
6	Laboratory + exercises	Trigger and explain failure modes
7	Benchmark + extension	Write one falsifiable experiment plan

Understanding means being able to predict both success and failure.

Mastery quiz: can you explain the whole pipeline?

01 Where do h_pq and g_pqrs come from? Which choices define the target?

02 What is the difference between the coherent span and the sampled basis?

03 Why does QKS need complex S and H matrix elements?

04 Why is S=I insufficient to guarantee an accurate SKQD energy?

05 Which inputs and classical controls would reproduce and test the water result?

The formula sheet: three core statements

01 **The target**

One specified active-space / sector eigenproblem.

$$H_{\text{target}} \left|E_0\right\rangle = E_0 \left|E_0\right\rangle$$

02 **Coherent QKS**

Transition amplitudes in a nonorthogonal time basis.

$$H_K c = E S_K c, \quad H_K = \Phi^\dagger H_{\text{target}} \Phi$$

03 **Sampled SKQD**

Target projection in an orthonormal sampled basis.

$$H_D = B_D^\dagger H_{\text{target}} B_D, \quad B_D^\dagger B_D = I$$

Primary reading: foundations

Label	Reference	Read it for
[M]	Ankush Kumar, uploaded manuscript, 8 Oct 2026	This specific implementation and reported results
[K]	Stair, Huang & Evangelista • arXiv:1911.05163	Multireference coherent quantum Krylov
[S]	Yu et al. • arXiv:2501.09702	Foundational sample-based Krylov method
[N]	Epperly, Lin & Nakatsukasa • arXiv:2110.07492	Conditioning and thresholded diagonalisation
[T]	Bravyi et al. • arXiv:1701.08213	Exact symmetry-sector tapering

Primary reading: related methods and extensions

Label	Reference	Read it for
[QSCI]	Kanno et al. • arXiv:2302.11320	Quantum-selected configuration interaction
[SQD]	Robledo-Moreno et al. • arXiv:2405.05068	Sample-based chemistry diagonalisation
[CI]	Holmes et al. • arXiv:1606.07453	Heat-bath selected CI and classical controls
[R]	Piccinelli et al. • arXiv:2508.02578	Randomized compilation for SKQD
[QITE]	Motta et al. • arXiv:1901.07653	Imaginary-time state updates

Your research task: a decisive controlled experiment

Choose a fixed target › **Reproduce the classical lab** › **Run SKQD and controls** › **Measure error & full cost**

01 Decide which conditions make sampled support useful.

02 Evaluate whether quantum discovery adds value beyond classical-only selection.

THE TAKEAWAY

Know the target.
Know the space.
Know the evidence.

Then build the next controlled experiment.